Sharmistha Jat

London, United Kingdom
+44 7404508846, sharmistha.jat@gmail.com

SUMMARY

Machine learning scientist with a bias to work in a fast-paced, research-driven environment, and contribute new ideas to solve complex problems. Self-starter, brings ideas to life with quick prototyping, ability to learn new domains quickly.

EDUCATION

PhD, Computer Science, 2015 - July 2021
Indian Institute of Science, Bengaluru (IISc), India
Advisors: Prof. Partha Talukdar (Indian Institute of Science-IISc, Bangalore) and Prof. Tom Mitchell (Carnegie Mellon University-CMU, Pittsburgh USA)
Thesis title: Relating Representations in Deep Neural Networks and the Brain
Pioneered methods to evaluate and compare representations in state-of-the-art LLMs against human brain activity, developing novel evaluation frameworks that provide insights into model performance. Designed and trained custom language models (`https://tinyurl.com/2tw5ry6d`), implementing innovative architectures that demonstrated improved performance on specific word classes. Published findings in top-tier conferences (ACL, NeurIPS, oHBM) and contributed to the broader scientific understanding of neural language models.

MTech, Computer Science, 2011-2013
Indian Institute of Technology Madras (IITM), Chennai, India
Advisor: Prof. Balaraman Ravindran
Thesis: Semi-Supervised Learning for Sentiment Analysis

WORK EXPERIENCE

Fintru, London, July 2023- current
Senior Machine Learning Scientist
Development, **fine-tuning** (RHLF/DPO), and deployment of LLM models for domain-specific information extraction tasks to improve model performance on key metrics. Develop, deploy, and maintain model cascades for latency and cost optimisations (quantization). Design and implement production-ready AI infrastructure that enables rapid iteration and deployment of models. Collaborate cross-functionally with engineering and product teams to ship novel AI applications that solve real business challenges. Experience working with the MLOps stack.

Spotify Ltd, London, Oct 2021- March 2023
Research Scientist
Developed novel methods for multi-modal (text and audio) content understanding, creating state-of-the-art embeddings that improved podcast genre prediction. Led research on hate speech detection and automated chapterization, shipping features that enhanced user experience for millions of listeners. Demonstrated scientific rigor through peer-reviewed publications while maintaining bias for action by rapidly prototyping and deploying solutions.

Xerox Research Centre India (XRCI), Bangalore, Oct 2013- July 2015
Research Engineer, Analytics Lab
Member of social media analytics and transportation research groups. Worked closely with Research Scientists to formulate and solve real-world problems using machine-learning techniques. Developed production-ready prototypes to integrate research tools into the business and published two papers and four patents

Morgan Stanley, Mumbai & New York, Jan 2010 - July 2011
Senior Associate
The role involved working with company wide data to maintain risk reserves. I helped design, develop, test and maintain end-to-end systems to process massive data streams. I worked actively in the frontend, middle-tier web services, and backend database management for the internal application.

INTERNSHIP EXPERIENCE

Carnegie Mellon University (CMU), Pittsburgh Sept 2017- April 2018, Sept 2018 - March 2019, Advisor: Prof. Tom Mitchell
Applied methodological precision in research, resulting in publications at top-tier conferences and received prestigious Brainhub doctoral fellowship to work on Ph.D. thesis. Collaborative Innovation: Worked effectively across disciplines (Neuroscience & Machine Learning) with researchers and data collection team (MEG brain scans) to deliver novel research at the intersection of computational neuroscience and NLP (ACL and OHBM publications)

TECHNICAL SKILLS

- Programming: Python (advanced), C/C++, Java, Matlab - ML Frameworks: PyTorch, TensorFlow, JAX - LLM Experience: Transformer architectures, fine-tuning techniques, prompt engineering, model evaluation - MLOps: Model evaluation frameworks, deployment pipelines, monitoring systems - Data Processing: NumPy, SciPy, Pandas, data preprocessing for LLMs - Cloud & - Databases: MySQL, MongoDB, vector databases for embeddings - Research: Publication record in top-tier conferences, patent development, cross-functional collaboration

PUBLICATIONS

Ekaterina Garmash, Edgar Tanaka, Ann Clifton, Joana Correia, Sharmistha Jat, Winstead Zhu and Rosie Jones and Jussi Karlgren, **Cem Mil Podcasts: A Spoken Portuguese Document Corpus for Multi-modal, Multi-lingual and Multi-dialect Information Access Research**, Accepted at International Conference of the CLEF Association, CLEF 2023, Thessaloniki, Greece 2023

Sharmistha Jat, Partha Talukdar, Tom Mitchell, **Word Salience during Sentence Reading**, Accepted at Human Brain Mapping **(oHBM)**, Seoul, Korea, 2021

Sharmistha Jat, Partha Talukdar, Tom Mitchell, **Empirical Evaluation of Composition in Deep Neural Networks using Brain Data** , 2021

Sharmistha Jat, Erika J C Laing, Partha Talukdar, Tom Mitchell, **Spatio-temporal Characteristics of Noun and Verb Processing during Sentence Comprehension in the Brain**, 2020, Bioarxiv

Sharmistha Jat, Hao Tang, Partha Talukdar, Tom Mitchell, **Relating Simple Sentence Representations in Deep Neural Networks and Brain**, Accepted at Association for Computational Linguistics **(ACL)**, Florence, 2019

S. Kumar, Sharmistha Jat, Karan Saxena, Partha Talukdar, **Zero-shot Word Sense Disambiguation using Sense Definition Embeddings**, Accepted at Association for Computational Linguistics **(ACL)**, Florence, 2019 [ACL outstanding paper award]

Sharmistha Jat, Tara Pirnia, Tom Mitchell, **Dynamic Time Warping for Brain MEG Analysis**, Accepted at Human Brain Mapping **(oHBM)**, Rome, 2019 [travel stipend award]

Nicole Rafidi, Mariya Toneva, Dan Schwartz, Sharmistha Jat, Tom Mitchell, **MEG Rep-**

resentational Similarity Analysis Implicates Hierarchical Integration in Sentence Processing, Organization for Human Brain Mapping, **OHBM** 2018, Singapore, June 17-21

Sharmistha Jat, Siddhesh Khandelwal and Partha Talukdar, **Improving Distantly Supervised Relation Extraction using Word and Entity Based Attention**, Automated Knowledge Base Construction Workshop, **NIPS 2017**, Long Beach, USA, Dec 4, 2017

Tushar Nagarajan, Sharmistha and Partha Talukdar, **CANDiS: Coupled & Attention-Driven Neural Distant Supervision**, The First Women and Underrepresented Minorities in NLP Workshop, **WiNLP, ACL 2017**, Vancouver, Canada, July 30, 2017

Lavanya Sita Tekumalla, Sharmistha, **IISCNLP at SemEval-2016 Task 2: Interpretable STS with ILP based Multiple Chunk Aligner**,workshop on Semantic Textual Similarity **(SemEval) NAACL 2016**, San Diego, 707-712

Himanshu Sharad Bhatt, Sandipan Dandapat, Peddamuthu Balaji, Shourya Roy, Sharmistha and Deepali Semwal, **SODA:Service Oriented Domain Adaptation Architecture for Microblog Categorization**, Proceedings of **NAACL 2016**, San Diego,77-81

Anuj Mahajan, Sharmistha, Shourya Roy, **Feature Selection for Short Text Classification using Wavelet Packet Transform**, Proceedings of the 19th Conference on Computational Natural Language Learning, **CoNLL 2015**, Beijing, China, July 30-31, 2015,ISBN - [978-1-941643-77-8], 321-326

Sharmistha, Madhur Amilkanthwar, Shankar Balachandran, **Augmentation of Programs with CUDA Streams**, 10th IEEE International Symposium on Parallel and Distributed Processing with Applications, **ISPA 2012**, Leganes, Madrid, Spain, July 10-13, 2012, ISBN - [978-1-4673-1631-6], 855-856

PATENTS GRANTED

Sharmistha Jat, Anuj Mahajan, Shourya Roy, **Method and System for Predicting Requirements of a User for Resources Over a Computer Network**, 2019, United States Patent 10417578

Sharmistha Jat, Koyel Mukherjee, Narayanan U. Edakunni, Pallavi Manohar, **Method and system for recommending one or more vehicles for one or more requesters**, 2018, USPTO Patent Number 9978111

Narayanan U. Edakunni, Sharmistha Jat, **Method and system for determining effect of weather conditions on transportation networks**, 2018, USPTO Patent Number 10062282

Koyel Mukherjee, Ankit Kumar, Pallavi Manohar, Narayanan U. Edakunni, Sharmistha Jat, **Method and system for scheduling vehicles along routes in a transportation system**, 2017, USPTO Patent Number 9746332

COURSES

Machine learning, Computational Methods of Optimisation, Stochastic Methods and Applications, Linear Algebra, Natural Language Understanding, Social Network Analysis, Knowledge Representation and Reasoning, Kernel Methods in Pattern Recognition

LINKS

`https://sharmisthajat.github.io/`
`https://www.linkedin.com/in/sharmistha-jat-126b9316/`
`https://github.com/SharmisthaJat`

https://scholar.google.co.in/citations?user=YVZ0NDwAAAAJ&hl=en&oi=ao

POSITION OF RESPONSIBILITY

Reviewer for NAACL 2024, ICLR 2021, CoNLL 2019, NIPS 2018, KDD 2018, XRCI Open Innovation Conference 2015

Teaching Assistant for Machine Learning (E0-270), Indian Institute of Science (IISc), January- April 2017

Tutorial, "Predictive Modeling Using R and Python" at Grace Hopper Conference (GHC), India, 8th December 2016

HONORS and AWARDS

BrainHub CMU-IISc Fellowship 2017-2018, 2018-2019

AAAI 2019 travel grant, OHBM 2019 travel stipend, NAACL 2016 travel grant

Top 6% in KAGGLE. Member of MALL Lab team which stood 10th among 170 teams for ALLEN-AI Kaggle challenge, demo: http://malllabiisc.github.io/demos/

Third position winner in Code-Obfuscation coding challenge at Shaastra, IIT Madras 2011

EXTRA-CURRICULAR ACTIVITIES

Member of XRCI Club 2013-2014, organised multiple charity and fun events.

Member of Charity Committee, Morgan Stanley 2010-2011.

Technology Volunteer for Commonwealth youth games, Pune 2008

Organized Girl Student Empowerment workshop at Department of CDS, IISc, with CHETANA Girls organisation.